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SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Java-Based Investment Portfolio Analysis, Optimization and Recommendation Framework Using Bayesian Networks

CSA0911 Programming in JAVA

Guided by
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ABSTRACT

- This project presents a Java-based framework that unifies stock, real estate, and precious-metal holdings into a single portfolio-tracking system for Indian investors.
- A Bayesian probability engine processes historical price, volatility, and fundamental and technical indicators to classify each asset as Bullish, Bearish, or Neutral and generate a Buy/Hold/Sell recommendation with a confidence score.
- The system is organized into four modules — data management, fundamental and technical analysis, Bayesian decision analysis, and visualization and reporting — culminating in an interactive dashboard and downloadable PDF reports.

INTRODUCTION

- Individual investors in India typically manage stocks, real estate, and gold or silver holdings through separate platforms, making it difficult to view overall portfolio risk and performance in one place.
- Most existing portfolio tools focus on displaying raw prices and valuations, offering little probabilistic or predictive insight to support buy, hold, or sell decisions.
- This project applies Bayesian networks alongside fundamental and technical analysis, built in Java with a modular architecture tailored to NSE stocks, Chennai real estate, and precious metals.

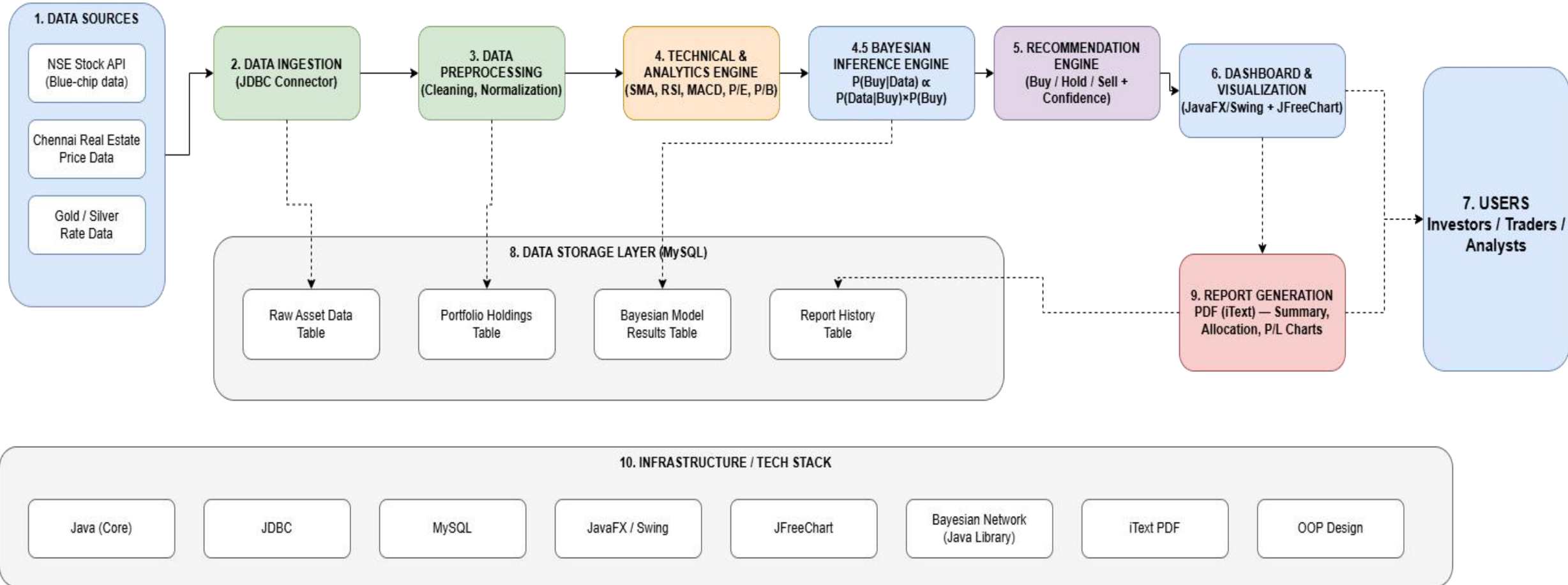
PROBLEM STATEMENT

- Investors track blue-chip stocks, real estate, and precious metals across disconnected platforms, with no single system unifying these asset classes or showing a clear market signal.
- Existing tools display raw prices and values but lack probabilistic risk analysis, leaving users without a data-driven Bullish/Bearish/Neutral indication to guide decisions.
- There is no lightweight, Java-based solution tailored to Indian market data — NSE blue-chips, Chennai real estate, and gold/silver — that combines tracking with predictive insight.

PROJECT SCOPE & OBJECTIVES

- Design a unified portfolio management system for stocks, real estate, and physical assets with manual entry, real-time value tracking, and risk scoring.
- Build a Bayesian recommendation engine that classifies each asset as Bullish, Bearish, or Neutral, and outputs a Buy/Hold/Sell call with a confidence score.
- Deliver an interactive dashboard with technical charts and downloadable PDF reports for portfolio summary, allocation, and profit/loss.

SYSTEM ARCHITECTURE



LITERATURE REVIEW

S.No	Research Paper	Author(s)	Advantages	Disadvantages	Technologies Used
1	From Classical Optimization to Bayesian Integration: A Comprehensive Analysis of Systematic Portfolio Management (2026)	Verma, A.K. & Barkam, S.	Compares Bayesian, mean-variance, and Black-Litterman models on real stock data; shows Bayesian integration improves allocation stability	Limited to 10 U.S. large-cap stocks; no real estate or gold asset class covered	Bayesian Modelling, Black-Litterman, Monte Carlo Simulation
2	A Comparative Study of Bayesian Portfolio Optimization: Evidence from U.S. Large-Cap AI-Related Stocks (2026)	Mikhailm, M.	Uses posterior sampling (2000 draws) to outperform Markowitz benchmark; strong empirical validation over 3-year training/1-year test split	High computational cost of posterior sampling; sector-specific	Bayesian Inference, Markowitz MVO, TradingView/Tiingo Data
3	Advanced Investing with Deep Learning for Risk-Aligned Portfolio Optimization (2025)	(PMC/NCBI published)	Combines LSTM forecasts with optimization strategies for better risk-adjusted returns; validated on 2023–2024 real market data	Deep learning models are compute-heavy and less interpretable than Bayesian probability output	LSTM, 1D-CNN, Mean-Variance Optimization
4	A Framework for Gold Price Prediction Combining Classical and Intelligent Methods with Financial, Economic, and Sentiment Data Fusion (2025)	(MDPI journal)	Fuses econometric, ML, and sentiment data for gold forecasting; validated on 11-year dataset across market conditions	Complex multi-model fusion increases system overhead for a lightweight app	Machine Learning, Sentiment Analysis, Econometric Modelling
5	Stock Recommendations for Individual Investors: A Temporal Graph Network Approach with Mean-Variance Efficient Sampling (2024)	Sanz-Cruzado, J. et al.	Tailors recommendations to individual investor behavior rather than generic market signals; introduces FAR-Trans dataset for finance recommendation research	Graph-based models need large transaction datasets; not designed for real-time dashboards	Temporal Graph Networks, Mean-Variance Sampling

MODULE 1: FINANCIAL DATA COLLECTION AND PORTFOLIO DATABASE MANAGEMENT

- Collect and organize financial data for Indian stocks, US stocks, mutual funds, ETFs, and real estate, including asset details, transactions, purchase price, quantity, and current value.
- Store and maintain five years of historical financial data for supported assets, creating a structured database for portfolio analysis and investment evaluation.
- Allow users to add, update, delete, and retrieve portfolio records, while maintaining accurate and organized information for each individual asset.
- Provide reliable financial and historical market data to the analysis, Bayesian processing, visualization, and reporting modules.

MODULE 1: FINANCIAL DATA COLLECTION AND PORTFOLIO DATABASE MANAGEMENT



CoinMarketCap API Pricing FAQ [Documentation](#)

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Basic	Hobbyist	Startup	Standard	Professional
Basic personal use	Personal projects	For commercial use	Suitable for most use cases	Best for scaling crypto projects
Free	\$29/mo	\$79/mo	\$299/mo	\$699/mo
No subscription required	\$420 \$348/year	\$1140 \$948/year	\$4500 \$3588/year	\$10500 \$8388/year
Get Your Free API Key	Get Started	Get Started	Get Started	Get Started
35+ data endpoints 15K call credits/mo - No historical data - Personal use	40+ data endpoints 150K call credits/mo 3 years of historical data - Personal use	50+ data endpoints 450K call credits/mo - All time historical data - Commercial use*	- All data endpoints 2M call credits/mo - All time historical data - Commercial use*	- All data endpoints 5M call credits/mo - All time historical data - Commercial use*

Enterprise [All data endpoints](#)

MODULE 2 – INTEGRATED FUNDAMENTAL AND TECHNICAL INVESTMENT ANALYSIS ENGINE

- Perform fundamental analysis using financial parameters such as revenue, profit, earnings, valuation ratios, and other relevant indicators to evaluate the financial strength of supported assets.
- Perform technical analysis using historical price and volume data with indicators such as moving averages, RSI, MACD, and other technical measures to identify market trends and patterns.
- Combine fundamental and technical analysis results to provide a comprehensive evaluation of individual assets and their potential investment performance.
- Generate structured analytical results from fundamental metrics, technical indicators, price trends, and historical data for use in Bayesian analysis and portfolio decision-making.

MODULE 2 – INTEGRATED FUNDAMENTAL AND TECHNICAL INVESTMENT ANALYSIS ENGINE

TechnicalAnalyzer.java — bayesian-portfolio — Visual Studio Code

File Edit Selection View Go Run Terminal Help

EXPLORER

- BAYESIAN-PORTFOLIO
 - src/main/java
 - com.portfolio.analysis
 - FundamentalAnalyzer.java
 - TechnicalAnalyzer.java
 - IndicatorEngine.java
 - AnalysisResult.java
 - com.portfolio.bayesian
 - ProbabilityEngine.java
 - RecommendationService.java
 - src/test/java
 - TechnicalAnalyzerTest.java
 - pom.xml
 - README.md

FundamentalAnalyzer.java **TechnicalAnalyzer.java** **IndicatorEngine.java**

```
41 // Module 2 – Fundamental & Technical Investment Analysis Engine
42 public class TechnicalAnalyzer {
43     private final List<PricePoint> history;
44
45     public TechnicalAnalyzer(List<PricePoint> history) {
46         this.history = history;
47     }
48
49     // Simple Moving Average over a given window
50     public double computeSMA(int window) {
51         return history.stream()
52             .skip(history.size() - window)
53             .mapToDouble(PricePoint::close)
54             .average().orElse(0.0);
55     }
56
57     // Relative Strength Index (14-day default)
58     public double computeRSI(int period) {
59         double gainSum = 0, lossSum = 0;
60         for (int i = history.size() - period; i < history.size(); i++) {
61             double delta = history.get(i).close() - history.get(i - 1).close();
62             if (delta >= 0) gainSum += delta; else lossSum -= delta;
63         }
64         double rs = (gainSum / period) / Math.max(lossSum / period, 1e-6);
65         return 100 - (100 / (1 + rs));
66     }
67 }
```

TERMINAL PROBLEMS OUTPUT DEBUG CONSOLE

```
PS C:\Users\ABDUL AZEEZ\bayesian-portfolio> mvn exec:java -Dexec.mainClass="com.portfolio.analysis.TechnicalAnalyzer"
[INFO] Loading price history for RELIANCE.NS (1258 records)
[INFO] SMA(50) = 2867.42 | SMA(200) = 2714.05
[INFO] RSI(14) = 58.7 | MACD = 12.34 | Signal = 9.81
[RESULT] Technical Bias: BULLISH — trend and momentum aligned
[BUILD SUCCESS] Total time: 4.102 s
```

Ln 66, Col 4 Spaces: 4 UTF-8 LF Java



MODULE 3 – BAYESIAN INVESTMENT PROBABILITY AND DECISION ANALYSIS ENGINE



- Apply Bayesian analysis to historical financial data and analytical results to estimate the probability of different investment outcomes for supported assets.
- Use historical returns, volatility, fundamental metrics, technical indicators, and other relevant parameters as evidence for calculating probability-based investment results.
- Generate investment indications such as Buy, Hold, or Sell based on the calculated Bayesian probabilities and available financial evidence.
- Present Bayesian probability results along with the underlying fundamental, technical, and historical information to provide transparent investment decision support.

MODULE 3 – BAYESIAN INVESTMENT PROBABILITY AND DECISION ANALYSIS ENGINE

ProbabilityEngine.java — bayesian-portfolio — Visual Studio Code

File Edit Selection View Go Run Terminal Help

EXPLORER

- ▼ BAYESIAN-PORTFOLIO
 - ▼ src/main/java
 - ▼ com.portfolio.analysis
 - FundamentalAnalyzer.java
 - TechnicalAnalyzer.java
 - IndicatorEngine.java
 - AnalysisResult.java
 - ▼ com.portfolio.bayesian
 - ProbabilityEngine.java
 - BayesianEvidence.java
 - RecommendationService.java
 - ▼ src/test/java
 - ProbabilityEngineTest.java
 - pom.xml
 - README.md

BayesianEvidence.java **ProbabilityEngine.java** **RecommendationService.java**

```
18 // Module 3 – Bayesian Investment Probability & Decision Analysis Engine
19 public class ProbabilityEngine {
20     private final List<Evidence> evidenceList;
21
22     public ProbabilityEngine(List<Evidence> evidenceList) {
23         this.evidenceList = evidenceList;
24     }
25
26     // P(Buy|Data) ∝ P(Data|Buy) × P(Buy) — Bayes' theorem
27     public double posteriorProbability(Signal signal) {
28         double prior = signal.getPrior();
29         double likelihood = evidenceList.stream()
30             .mapToDouble(e -> e.likelihoodGiven(signal))
31             .reduce(1.0, (a, b) -> a * b);
32         return normalize(likelihood * prior);
33     }
34
35     // Classify asset from posterior probabilities across all signals
```

TERMINAL PROBLEMS OUTPUT DEBUG CONSOLE

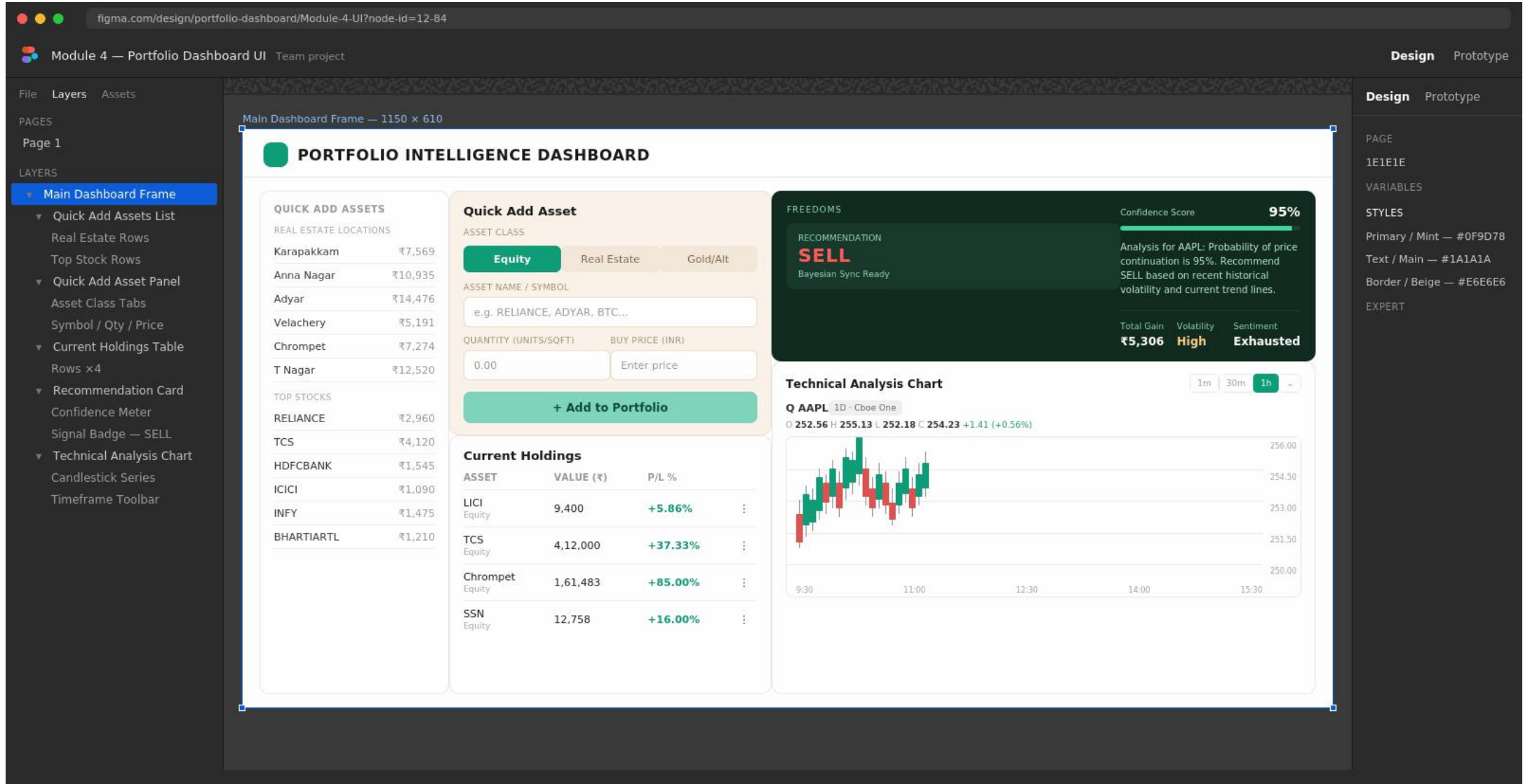
```
PS C:\Users\ABDUL AZEEZ\bayesian-portfolio> mvn exec:java -
Dexec.mainClass="com.portfolio.bayesian.ProbabilityEngine"
[INFO] Loading evidence: fundamentals, technicals, volatility, historical returns
[INFO] P(Buy|Data) = 0.71 | P(Hold|Data) = 0.21 | P(Sell|Data) = 0.08
[INFO] Evaluating RELIANCE.NS against posterior distribution
[RESULT] Recommendation: BUY — confidence 71.0%
[BUILD SUCCESS] Total time: 3.687 s
```

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MODULE 4 – INTERACTIVE FINANCIAL VISUALIZATION AND PORTFOLIO USER INTERFACE

- Provide an interactive user interface for viewing portfolio holdings, investment value, returns, profit or loss, asset allocation, and analysis results in a structured manner.
- Display financial information through interactive charts, graphs, tables, and performance indicators to simplify the interpretation of complex investment data.
- Visualize historical trends, fundamental metrics, technical indicators, portfolio performance, and Bayesian results for effective comparison across different assets.
- Provide downloadable financial reports combining portfolio information, analytical results, visualizations, and Bayesian insights into a consolidated investment report.

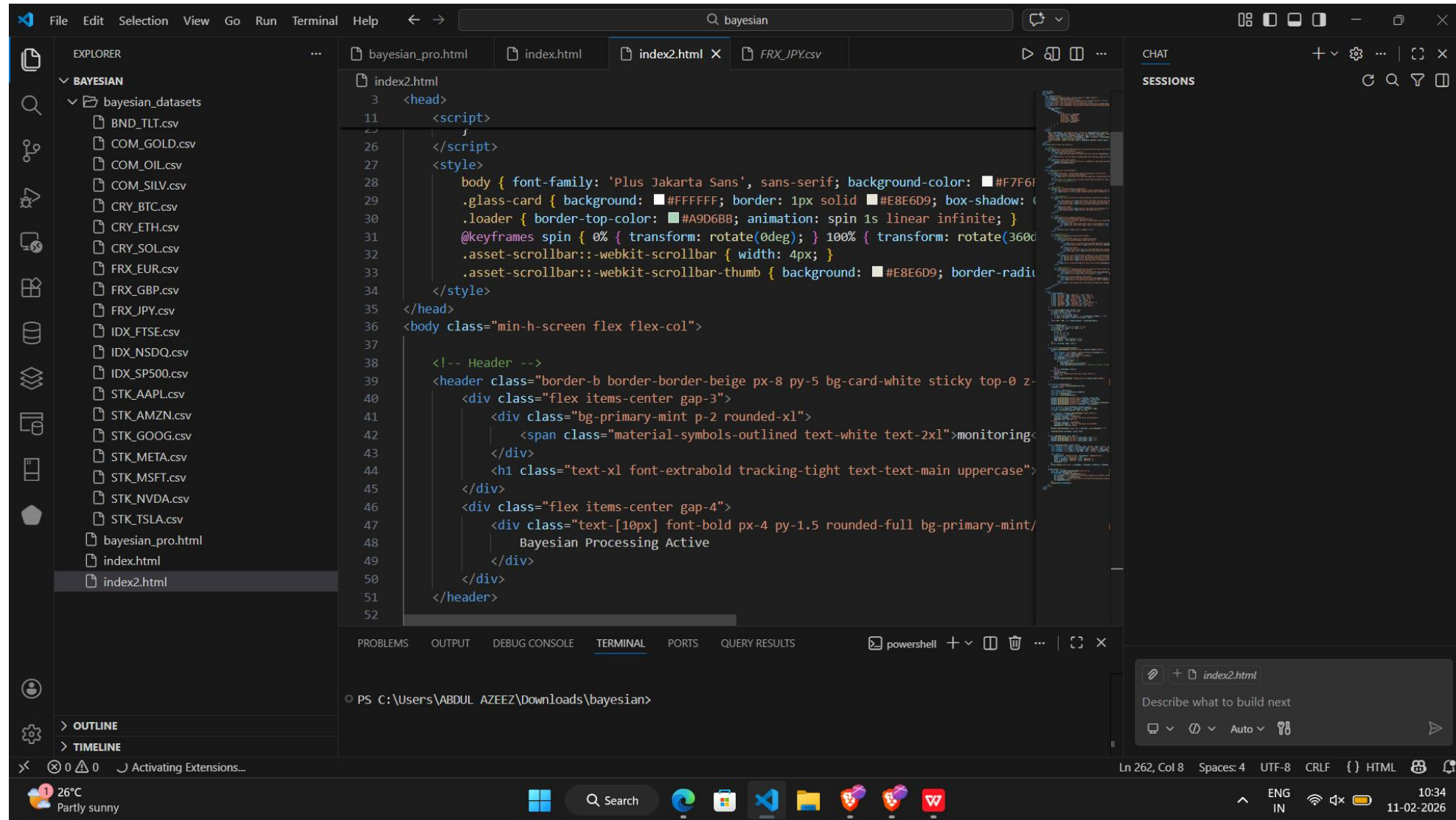
MODULE 4 – INTERACTIVE FINANCIAL VISUALIZATION AND PORTFOLIO USER INTERFACE



EVALUATION METRICS

- Technical Indicators — SMA 50, SMA 200, RSI, and MACD are calculated for each asset and checked against actual price charts for correctness.
- Bayesian Formula Accuracy — the posterior probability calculation ($P(\text{Buy}|\text{Data}) \propto P(\text{Data}|\text{Buy}) \times P(\text{Buy})$) is validated by comparing predicted sentiment against real market movement.
- Volatility Accuracy — standard deviation of daily returns (annualized volatility %) is compared against actual historical price fluctuation for each asset.
- Fundamental Ratios — P/E Ratio, P/B Ratio, and ROI are cross-checked against real company financials to confirm the accuracy of displayed valuation metrics.

SIMULATION RESULTS

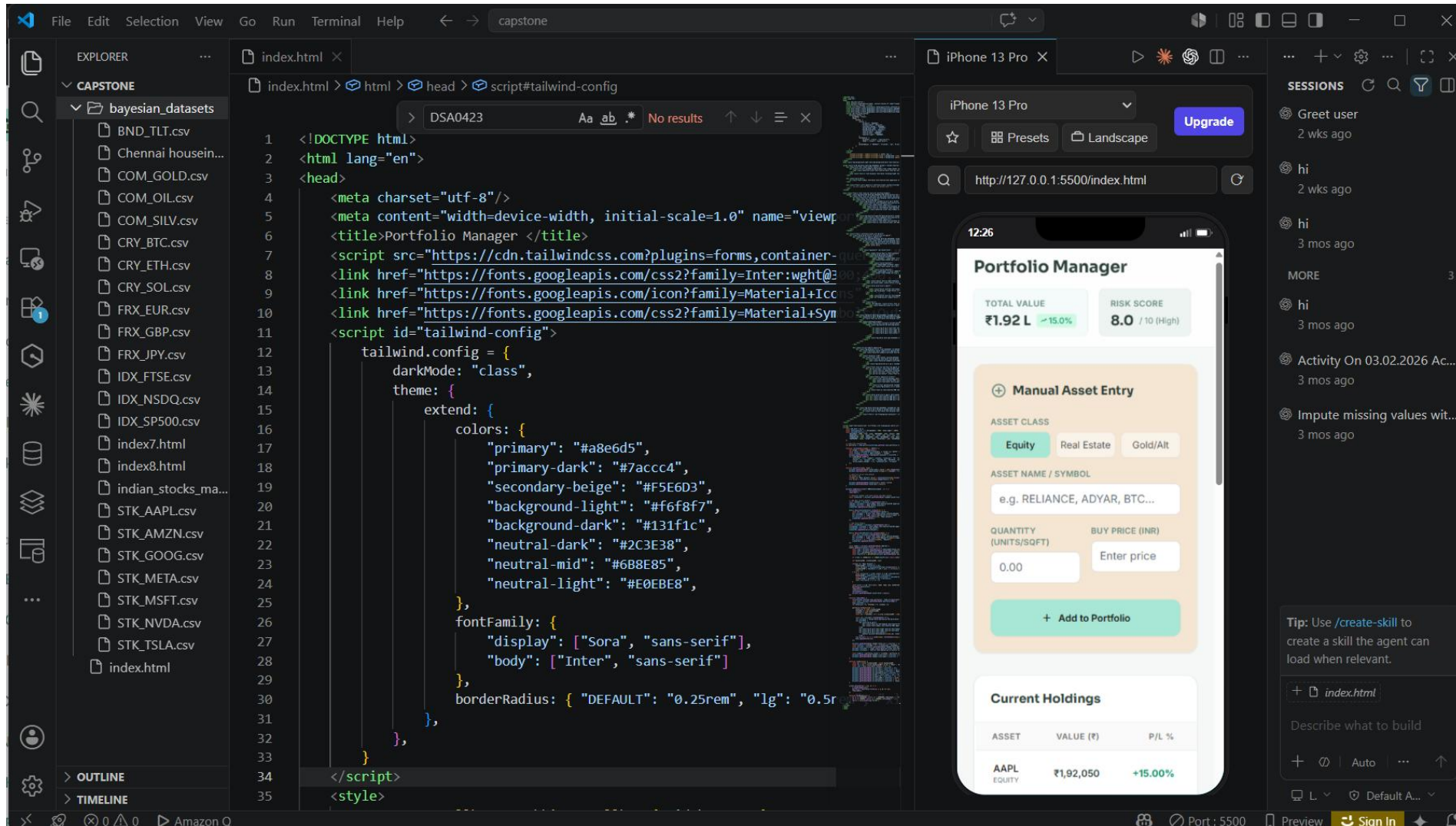


The screenshot displays the Visual Studio Code (VS Code) editor interface. The Explorer panel on the left shows a project named 'BAYESIAN' with a subfolder 'bayesian_datasets' containing various CSV files (e.g., BND_TLT.csv, COM_GOLD.csv, COM_OIL.csv, COM_SILV.csv, CRY_BTC.csv, CRY_ETH.csv, CRY_SOL.csv, FRX_EUR.csv, FRX_GBP.csv, FRX_JPY.csv, IDX_FTSE.csv, IDX_NSDQ.csv, IDX_SP500.csv, STK_AAPL.csv, STK_AMZN.csv, STK_GOOG.csv, STK_META.csv, STK_MSFT.csv, STK_NVDA.csv, STK_TSLA.csv). The main editor area shows the 'index2.html' file, which contains HTML and CSS code. The code includes a script, a style block, and a body section with a header and a main content area. The header section includes a 'monitoring' label and a 'Bayesian Processing Active' status. The main content area is a large, empty div. The bottom status bar shows the file path 'PS C:\Users\ABDUL AZEEZ\Downloads\bayesian' and the terminal output 'PS C:\Users\ABDUL AZEEZ\Downloads\bayesian'.

```
<script>
</script>
<style>
  body { font-family: 'Plus Jakarta Sans', sans-serif; background-color: #F7F6F6; }
  .glass-card { background: #FFFFFF; border: 1px solid #E8E6D9; box-shadow: 0 4px 6px #E8E6D9; }
  .loader { border-top-color: #A9D6BB; animation: spin 1s linear infinite; }
  @keyframes spin { 0% { transform: rotate(0deg); } 100% { transform: rotate(360deg); } }
  .asset-scrollbar::-webkit-scrollbar { width: 4px; }
  .asset-scrollbar::-webkit-scrollbar-thumb { background: #E8E6D9; border-radius: 2px; }
</style>
</head>
<body class="min-h-screen flex flex-col">

  <!-- Header -->
  <header class="border-b border-border-beige px-8 py-5 bg-card-white sticky top-0 z-10">
    <div class="flex items-center gap-3">
      <div class="bg-primary-mint p-2 rounded-xl">
        <span class="material-symbols-outlined text-white text-2xl">monitoring</span>
      </div>
      <h1 class="text-xl font-extrabold tracking-tight text-text-main uppercase">MONITORING</h1>
    </div>
    <div class="flex items-center gap-4">
      <div class="text-[10px] font-bold px-4 py-1.5 rounded-full bg-primary-mint/50">
        Bayesian Processing Active
      </div>
    </div>
  </div>
</header>
```

SIMULATION RESULTS



The screenshot displays a web development environment with the following components:

- EXPLORER:** A file tree showing a project named 'CAPSTONE' with a subfolder 'bayesian_datasets' containing various CSV files (e.g., BND_TLT.csv, Chennai housein..., COM_GOLD.csv, etc.) and an 'index.html' file.
- Code Editor:** Displays the 'index.html' file with the following content:

```
1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="utf-8"/>
5   <meta content="width=device-width, initial-scale=1.0" name="viewport"/>
6   <title>Portfolio Manager </title>
7   <script src="https://cdn.tailwindcss.com?plugins=forms,container-queries"></script>
8   <link href="https://fonts.googleapis.com/css2?family=Inter:wght@300;400;500;600;700;800;900&display=block" rel="stylesheet">
9   <link href="https://fonts.googleapis.com/icon?family=Material+Icons">
10  <link href="https://fonts.googleapis.com/css2?family=Material+Symbols+Outlined" rel="stylesheet">
11  <script id="tailwind-config">
12    tailwind.config = {
13      darkMode: "class",
14      theme: {
15        extend: {
16          colors: {
17            "primary": "#a8e6d5",
18            "primary-dark": "#7acc4",
19            "secondary-beige": "#F5E6D3",
20            "background-light": "#f6f8f7",
21            "background-dark": "#131f1c",
22            "neutral-dark": "#2C3E38",
23            "neutral-mid": "#6B8E85",
24            "neutral-light": "#E0EBE8",
25          },
26          fontFamily: {
27            "display": ["Sora", "sans-serif"],
28            "body": ["Inter", "sans-serif"]
29          },
30          borderRadius: { "DEFAULT": "0.25rem", "lg": "0.5rem" }
31        }
32      }
33    }
34  </script>
35  <style>
```
- Preview:** A mobile preview of the 'Portfolio Manager' app on an iPhone 13 Pro. The app interface includes:
 - Header:** 'Portfolio Manager' title.
 - Summary:** 'TOTAL VALUE ₹1.92 L' (up 15.0%) and 'RISK SCORE 8.0 / 10 (High)'.
 - Manual Asset Entry:** A form with 'ASSET CLASS' (Equity, Real Estate, Gold/Alt), 'ASSET NAME / SYMBOL' (e.g., RELIANCE, ADYAR, BTC...), 'QUANTITY (UNITS/SQFT)' (0.00), and 'BUY PRICE (INR)' (Enter price). A '+ Add to Portfolio' button is at the bottom.
 - Current Holdings:** A table with columns 'ASSET', 'VALUE (₹)', and 'P/L %'. It shows 'AAPL EQUITY' with a value of ₹1,92,050 and a P/L of +15.00%.
- SESSIONS:** A list of sessions on the right side, including 'Greet user' (2 wks ago), 'hi' (2 wks ago), 'hi' (3 mos ago), 'Activity On 03.02.2026 Ac...' (3 mos ago), and 'Impute missing values wit...' (3 mos ago).
- Footer:** A 'Tip: Use /create-skill to create a skill the agent can load when relevant.' and a 'Describe what to build' section with a '+ index.html' button and a 'Describe what to build' input field.

SIMULATION RESULTS

QUICK ADD ASSETS

REAL ESTATE LOCATIONS

Karapakkam	₹7569
Anna Nagar	₹10935
Adyar	₹14476
Velachery	₹5191
Chrompet	₹7274
Kk Nagar	₹5435
T Nagar	₹12530

TOP STOCKS

RELIANCE	₹2980
TCS	₹4120
HDFCBANK	₹1545
ICICI	₹1090
INFY	₹1475
BHARTIARTL	₹1210
SBIN	₹775

ASSET CLASS

Equity

Real Estate

Gold/Alt

ASSET NAME / SYMBOL

e.g. RELIANCE, ADYAR, BTC...

QUANTITY (UNITS/SQFT)

0.00

BUY PRICE (INR)

Enter price

+ Add to Portfolio

Current Holdings

ASSET	VALUE (₹)	P/L %	ACTION
LICI EQUITY	₹9,400	+5.86%	
TCS EQUITY	₹4,12,000	+37.33%	
CHROMPET	₹1.61.483	+85.00%	

RECOMMENDATION

SELL

Bayesian Sync Ready

Confidence Score 93%

Analysis for AAPL: Probability of price continuation is 93%. Recommend SELL based on recent historical volatility and current trend lines.

Total Gain

₹2,505

Volatility

High

Sentiment

Exhausted

Technical Analysis Chart



FUTURE SCOPE

- Extend asset coverage to mutual funds, ETFs, and cryptocurrency, building on the existing modular data and analysis architecture.
- Incorporate news and sentiment data alongside fundamental and technical indicators to further refine the Bayesian probability estimates.
- Add real-time market data feeds, automated alerts, and a mobile companion app so recommendations stay current and are accessible on the go.

CONCLUSION

- The project bridges the gap between passive asset tracking and active, data-driven decision-making by unifying stocks, real estate, and metals into one dashboard.
- Bayesian modelling adds a probabilistic edge over conventional tracking tools, converting raw price and volatility data into a confidence-scored Buy/Hold/Sell signal.
- The modular architecture (Data, Analytics, Recommendation, Reporting) ensures scalability for future asset classes, allowing sources like mutual funds or crypto to be added later.
- This system empowers everyday investors in India with an accessible, intelligent portfolio tool, reflecting practical use of databases and probabilistic modelling learned through the course.

THANK YOU

PROOF

